

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	2024	Date:	01/08/2026

Code of Conduct

I _____ C Prathibha _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ C Prathibha _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

4. A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.

Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.

Analyze how prefix addition and suffix transformation affect meaning and word class.

Apply the process to decompose each word and map them to a unified base representation.

Derive the morphological structure and normalized root for each input word.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.

Identify suffix patterns and map all variations to a common base form without altering meaning.

Apply the process to extract root forms and classify each transformation.

Determine the morphological breakdown and normalized output for each word.

ANSWERS

1. Search Engine Indexing System

Input Words: analyzing, analysis, analytical

Morphological Processing Approach

- Identify the base/root word.
- Separate prefixes and suffixes using morphological rules.
- Classify the word as inflectional or derivational.
- Normalize all words to a common base form for indexing.

Morphological Breakdown

Word	Root	Affix	Type	Normalized Form
analyzing	analyze	-ing	Inflectional	analyze
analysis	analyze	-sis	Derivational	analyze

Word	Root	Affix	Type	Normalized Form
analytical	analyze	-ical	Derivational	analyze

Morphological Structure

analyzing

Root : analyze

Suffix : -ing

analysis

Root : analyze

Suffix : -sis

analytical

Root : analyze

Suffix : -ical

Final Output

All three words are normalized to the common base form:

Normalized Root = analyze

2. AI-Based Sentiment Analysis

Input Words: disagree, agreement, agreeable

Morphological Parsing Strategy

- Identify prefixes and suffixes.
- Extract the common base form.
- Classify each transformation.
- Preserve semantic meaning for sentiment analysis.
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Morphological Breakdown

Word	Prefix	Root	Suffix	Type
disagree	dis-	agree	—	Derivational
agreement	—	agree	-ment	Derivational
agreeable	—	agree	-able	Derivational

Meaning Change

- dis- → expresses negation.
- -ment → forms a noun.
- -able → forms an adjective.

Morphological Structure

disagree

Prefix : dis-

Root : agree

agreement

Root : agree

Suffix : -ment

agreeable

Root : agree

Suffix : -able

Final Output

Word	Normalized Root
disagree	agree
agreement	agree
agreeable	agree

3. Text Analytics Pipeline

Input Words: govern, government, governance

Morphology-Based Normalization

- Detect root word.
- Remove derivational suffixes.
- Normalize all forms to the same base.

Morphological Breakdown

Word	Root	Suffix	Type	Normalized Form
govern	govern	—	Base Word	govern
government	govern	-ment	Derivational	govern
governance	govern	-ance	Derivational	govern

Morphological Structure

govern

Root : govern

government

Root : govern

Suffix : -ment

governance

Root : govern

Suffix : -ance

Final Output

Input	Normalized Root
govern	govern
government	govern
governance	govern

4. Document Classification System

Input Words: activate, activation, reactivation

Morphological Parsing Strategy

- Identify prefixes and suffixes.
- Detect derivational layers.
- Map all words to one base representation.

Morphological Breakdown

Word	Prefix	Root	Suffix	Type
activate	—	activate	—	Base Word

Word	Prefix	Root	Suffix	Type
activation	—	activate	-ion	Derivational
reactivation	re-	activate	-ion	Derivational

Meaning Change

- re- → means "again".
- -ion → changes verb into noun.

Morphological Structure

activate

Root : activate

activation

Root : activate

Suffix : -ion

reactivation

Prefix : re-

Root : activate

Suffix : -ion

Final Output

Word	Normalized Root
activate	activate
activation	activate
reactivation	activate

5. Search Optimization Module

Input Words: create, creates, creating

Morphology-Based Normalization

- Detect inflectional suffixes.
- Remove tense and aspect markers.
- Normalize all words to the base form.

Morphological Breakdown

Word	Root	Suffix	Type	Normalized Form
create	create	—	Base Word	create
creates	create	-s	Inflectional	create
creating	create	-ing	Inflectional	create

Morphological Structure

create

Root : create

creates

Root : create

Suffix : -s

creating

Root : create

Suffix : -ing

Final Output

Word	Transformation	Normalized Root
create	Base form	create
creates	Present tense (Inflectional)	create
creating	Present participle (Inflectional)	create